SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3 19/20 **Total Marks:30**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:
I Abarajithan S I 192521299 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S.Ab

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

1. Remote rural branch office

Recommended internet connection:

Use a 4G/5G fixed wireless connection as the primary connection, If 5G/4G coverage is weak, use VSAT satellite internet. For uninterrupted operation, a dual-WAN router can automatically switch between the primary and backup links.

Networking devices:

—> 4G/5G modem or satellite terminal

—> Dual-WAN business router.

—> Gigabit Ethernet switch.

—> Wifi 6 access points.

—> VPN-capable firewall.

—> UPS for Network equipment.

Security mechanisms:

—> Site-to-site IPsec VPN between branch and headquarters.

—> Firewall with intrusion prevention.

—> WPA3 for wifi.

—> Strong user authentication.

—> VLANs to separate employees, guests, and critical systems.

—> Regular firmware and security updat

Justification:

—> Bandwidth: 4G/5G can provide sufficient bandwidth for cloud application and video meetings where coverage is good

—> Reliability: A backup satellite/4G connection provides connectivity if the primary link fails.

—> Cost: Fixed wireless generally costs less to deploy than installing new wired infrastructure in a rural are

—> Security: VPN encryption protect communication with headquarters.

Best solution:

—> 4G/5G + satellite/secondary WAN + Firewall + VPN + VLANs provides a good balance between cost, reliability security, and scalability.

2. University wireless Network for 3,000+ students

Network design:

→ The university should use a centrally managed wifi 6 / Wifi 6E network with carefully positioned access points rather than simply installing APs everywhere

Access Point placement:

→ Place APs in classrooms based on expected student density.

→ Install APs in libraries and lab where simultaneous usage is high.

→ provide APs in hostel common areas and corridors.

→ perform a wireless site survey before final installation.

→ Avoid placing APs too close together because excessive overlap causes interference.

→ Use directional antennas where appropriate in large halls.

Efficient design:

⟶ Instead of providing maximum signal strength everywhere, design for capacity and controlled coverage. High density areas should receive more carefully positioned APs, while low-density areas can use fewer APs.

Justification:

⟶ coverage: Strategic AP Placement covers classrooms, hostels, laboratories and libraries.

⟶ performance: 5/6 GHz and wifi 6 improve capacity for many simultaneous users.

⟶ Security: WPA3-Enterprise and 802.1x provide individual authentication

⟶ Interference: channel planning and reduced transmit power prevent excessive AP overlap.

⟶ cost: limited APs are used efficiently by prioritizing high-density locations.

3. Secure Network for a Financial institution

Proposed Architecture:

Use a segmented, layered network architecture with firewalls and controlled access between network zones.

Network Segmentation:

Create separate VLANS/Security zones for:

→ Employee computers.

→ customer-information systems.

→ Banking/applications servers.

→ Database servers.

→ Guest users.

→ Management systems.

Sensitive customer databases should never be directly accessible from normal employee networks.

Authentication:

→ Use multi-factor authentication (MFA) for sensitive applications and administrative access.

→ Implement role-based access control (RBAC)

→ Use centralized identity management

→ Follow the principle of least privilege.

Intrusion Detection and Prevention:

→ Deploy IDS/IPS to monitor suspicious network traffic.

→ Use centralized logging and security monitoring/SIEM.

→ Monitor failed login attempts, unusual data transfer and unauthorized access.

→ Regularly update firewall, IDS/IPS and end point security signatures.

Justification:

constraint	Solution.
security	VLANs, firewalls, MFA, IDS/IPS and least-privilege access
Performance	separate high-traffic networks and use efficient firewall rules.
Budget	use centralized security management & existing network infrastructure
compliance	strong authentication, access control, logging and monitoring.